



Part 1: Multiple choice questions

Q1: The linear system

$$x + y - z = 2$$

$$x + 2y + z = 3$$

$$x + y + (a^2 - 5)z = a$$

has unique solution if $a \in$

A) $\mathbb{R} - \{-2, 2\}$

B) $\{-2\}$

C) $\{2\}$

D) $\{-2, 2\}$

Q2: Let $u = (2, 0, 0)$. Then the value(s) of the scalar k such that $\|ku\| = 20$ is

A) $\mathbb{R} - \{-6, 6\}$

B) $\{-4\}$

C) $\{5\}$

D) $\{-5, 5\}$

Q3: Let $\det(A) = 2$ and $\det(B) = -2$, A and B are 2×2 matrices. Then the value of $\det(2A^t B^{-1})$ is:

A) 4

B) -4

C) 16

D) -16

Q4: If $b \neq 0$

$$A = \begin{bmatrix} b & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & b \end{bmatrix}$$

$$B = \begin{bmatrix} -3 & 2 & 1 \\ 0 & b & 1 \\ 0 & 0 & b \end{bmatrix}$$

then $\det(2AB^{-1}) =$

A) 8

B) -8

C) 1

D) -1

Q5: The value of k that makes $\begin{pmatrix} k & 2k \\ 1 & k \end{pmatrix}$ singular is:

A) $\{0, 3\}$

B) $\{0, 2\}$

C) 3

D) 2

Q6: If $A = \begin{bmatrix} c & 2 & 0 \\ 0 & c & 1 \\ 0 & 0 & c \end{bmatrix}$, and $\det(2A) = 64$, then the value of $2c =$

A) 2

B) 4

C) 6

D) 8

Q7: If $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 1 \end{bmatrix}$, then $M_{22} =$

A) 1

B) -1

C) 3

D) -3

Q8: If $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 2 \\ 2 & 1 & 1 \end{bmatrix}$, then $C_{22} =$

A) 1

B) -1

C) 3

D) -3

$$2^3 \times 2^3$$

$$1 - 2 = -1$$

$$1 - 2 = (-1)$$

$$k - 2k = 0$$

$$k(1 - 2) = 0$$

$$k(-1) = 0$$

$$k = 0$$

$$(1) (1) (1)$$



Part two: True or False questions:

Q1: The transpose of an upper triangular matrix is an upper triangular.

~~(TRUE)~~ (FALSE)
(TRUE) ~~(FALSE)~~

Q2: $\det(A + B) = \det(A) + \det(B)$.

Q3: If $\det(2A) = 0$, for a square matrix of order n , then A is singular (not invertible) matrix.

(TRUE) ~~(FALSE)~~

Q4: If the dot product of two vectors equals zero, then one of the vectors is the zero vector.

(TRUE) ~~(FALSE)~~

$$2A = 0$$

$$2^n \det A$$

$$2 \det A$$

$$\det(2A^t B^{-1})$$

$$-2) = 2/0$$

$$0) \neq 1$$



Part three: Written questions. (6 marks)

Solve the following linear system of equations using Cramer's rule.

$$x + y = z + 0$$

$$y + z = -3 \quad y = -3 - z$$

$$x = z + 1$$

$$x - z = 1$$

Solution:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 1 & -3 \\ 1 & 0 & -1 & 1 \end{array} \right]$$

$$x_1 = \frac{\det A_1}{\det A}$$

$$= \frac{\begin{vmatrix} 0 & 1 & -1 \\ -3 & 1 & 1 \\ 1 & 0 & -1 \end{vmatrix}}{1} = \frac{-1}{1}$$

$$x_2 = \frac{\det A_2}{\det A}$$

$$= \frac{\begin{vmatrix} 1 & 0 & -1 \\ 0 & -3 & 1 \\ 1 & 1 & -1 \end{vmatrix}}{1} = \frac{-1}{1}$$

$$x_3 = \frac{\det A_3}{\det A}$$

$$= \frac{\begin{vmatrix} 1 & 1 & 0 \\ 0 & 1 & -3 \\ 1 & 0 & 1 \end{vmatrix}}{1} = \frac{-2}{1}$$

$$AX = B$$

$$X = A^{-1}B$$

$$B = \begin{bmatrix} 0 \\ -3 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\det = -1 \neq 1$$

$$(1)(2) - (0)(1) + (-1)(1)$$

$$\det A = 1$$

$$x_1 = -1 \quad x_2 = -1$$

$$x_3 = -2$$